Topic 3: Redox I

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include simple test tube reactions to investigate redox systems.

Mathematical skills that could be developed in this topic include using an algebraic method to work out the oxidation number of an element within a complex species, balancing equations for redox reactions by combining ionic half-equations.

Within this topic, students can consider how the concept of oxidation number provides a more considered route for the process of balancing chemical equations.

Students should:	
1.	know what is meant by the term 'oxidation number'
2.	be able to calculate the oxidation number of elements in compounds and ions *The use of oxidation numbers in peroxides and metal hydrides is expected.*
3.	understand oxidation and reduction in terms of electron transfer and changes in oxidation number, applied to reactions of *s*- and *p*-block elements
4.	understand oxidation and reduction in terms of electron loss or electron gain
5.	know that oxidising agents gain electrons
6.	know that reducing agents lose electrons
7.	understand that a disproportionation reaction involves an element in a single species being simultaneously oxidised and reduced
8.	know that oxidation number is a useful concept in terms of the classification of reactions as redox and as disproportionation
9.	be able to indicate the oxidation number of an element in a compound or ion, using a Roman numeral
10.	be able to write formulae given oxidation numbers
11.	understand that metals, in general, form positive ions by loss of electrons with an increase in oxidation number
12.	understand that non-metals, in general, form negative ions by gain of electrons with a decrease in oxidation number
13.	be able to write ionic half-equations and use them to construct full ionic equations

Topic 4: Inorganic Chemistry and the Periodic Table

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include reacting Group 2 elements with water, heating nitrates and carbonates of Group 1 and 2 elements, investigating flame colours of *s*-block elements, preparing iodine from seaweed, investigating displacement reactions in the halogens, reacting Group 1 halides with concentrated sulfuric acid.

Mathematical skills that could be developed in this topic include manipulating data on the solubility of hydroxides.

Within this topic, students can consider how data can be used to make predictions based on patterns and relationships, for example by predicting properties of Group 7 elements.